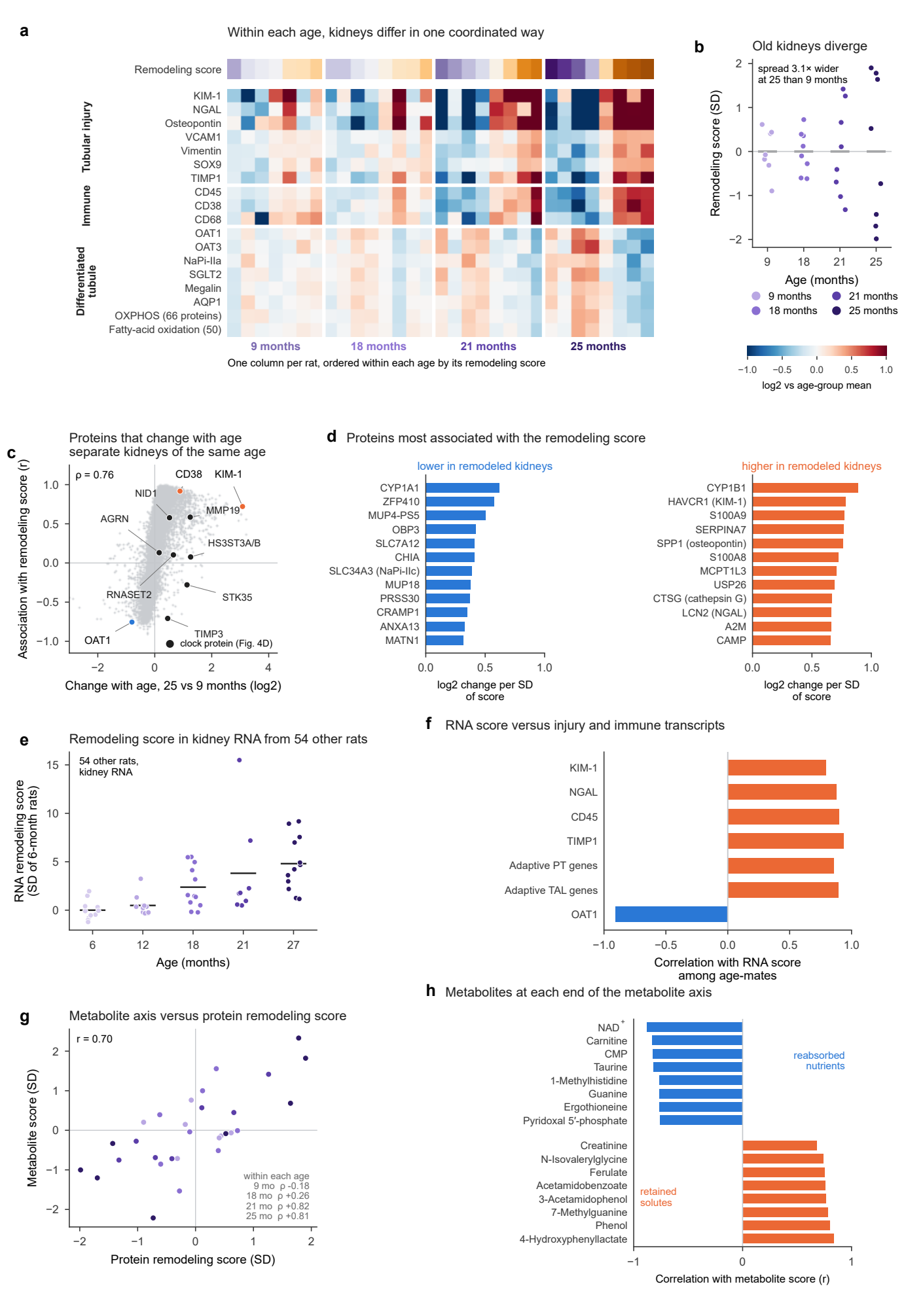

a
Within each age, kidneys differ in one coordinated way
Remodeling score
Tubular injury
KIM-1
NGAL
Osteopontin
VCAM1
Vimentin
SOX9
TIMP1
Immune
CD45
CD38
CD68
Differentiated tubule
OAT1
OAT3
NaPi-IIa
SGLT2
Megalin
AQP1
OXPHOS (66 proteins)
Fatty-acid oxidation (50)
9 months
18 months
21 months
25 months
One column per rat, ordered within each age by its remodeling score
b
Old kidneys diverge
spread 3.1× wider at 25 than 9 months
Remodeling score (SD)
Age (months)
9 months
18 months
21 months
25 months
log2 vs age-group mean
c
Proteins that change with age separate kidneys of the same age
ρ = 0.76
CD38
KIM-1
NID1
MMP19
AGRN
HS3ST3A/B
RNASET2
STK35
OAT1
TIMP3
clock protein (Fig. 4D)
Association with remodeling score (r)
Change with age, 25 vs 9 months (log2)
d
Proteins most associated with the remodeling score
lower in remodeled kidneys
CYP1A1
ZFP410
MUP4-PS5
OBP3
SLC7A12
CHIA
SLC34A3 (NaPi-IIc)
MUP18
PRSS30
CRAMP1
ANXA13
MATN1
log2 change per SD of score
higher in remodeled kidneys
CYP1B1
HAVCR1 (KIM-1)
S100A9
SERPINA7
SPP1 (osteopontin)
S100A8
MCPT1L3
USP26
CTSG (cathepsin G)
LCN2 (NGAL)
A2M
CAMP
log2 change per SD of score
e
Remodeling score in kidney RNA from 54 other rats
54 other rats, kidney RNA
RNA remodeling score (SD of 6-month rats)
Age (months)
f
RNA score versus injury and immune transcripts
KIM-1
NGAL
CD45
TIMP1
Adaptive PT genes
Adaptive TAL genes
OAT1
Correlation with RNA score among age-mates
g
Metabolite axis versus protein remodeling score
r = 0.70
within each age
9 mo ρ -0.18
18 mo ρ +0.26
21 mo ρ +0.82
25 mo ρ +0.81
Metabolite score (SD)
Protein remodeling score (SD)
h
Metabolites at each end of the metabolite axis
NAD+
Carnitine
CMP
Taurine
1-Methylhistidine
Guanine
Ergothioneine
Pyridoxal 5′-phosphate
reabsorbed nutrients
Creatinine
N-Isovalerylglycine
Ferulate
Acetamidobenzoate
3-Acetamidophenol
7-Methylguanine
Phenol
4-Hydroxyphenyllactate
retained solutes
Correlation with metabolite score (r)